1 Application Documents

The examination is being carried out on the **following application documents**:

Description, Pages

1-14 as published

Claims, Numbers

1-17 as published

Drawings, Sheets

1/4-4/4 as published

2 Cited Documents

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 4 295 535 A (CRASE GARY M ET AL) 20 October 1981 (1981-10-20)

D2 GB 2 055 927 A (ENG ENTERPR) 11 March 1981 (1981-03-11)

D3 US 6 241 032 B1 (FALGOUT SR THOMAS E [US]) 5 June 2001 (2001-06-05)

3 Clarity and Conciseness (Art. 84 EPC)

3.1 The application does not meet the requirements of Article 84 EPC, because claims 3, 5, 9, 11, 12, 13, 14, 15 are not clear.

3.2 The sentence "the tubular housing 104 is removably attached to a power output shaft of a downhole drilling motor" used in claim 3 is unclear. It appears from the drawings and description (§12) that "the tubular housing 104" should in fact be replaced by "the output drive shaft 102".

The same remark applies to claim 15.

3.3 The feature "the ratchet teeth" used in claim 5 has not been previously defined.

3.4 The feature "a third rotational speed" used in claim 9 is unclear as no "second rotational speed" has been previously defined.

The same remark applies to claim 14.

3.5 The feature "longitudinal grooves on the output drive shaft" used in claim 11 is not clear. It appears from the drawings and description (§15) that the longitudinal grooves are located on the internal wall of the bore of the driven gear.

3.6 The feature "the grooves of the driving gear" used in claim 12 is not clear. It appears (see above) that these grooves pertain to the driven gear.

3.7 The sentence "rotating the output shaft [...] in the first directional direction urges the ball-end screw" used in claim 13 is not grammatically correct.

The same remark applies to claim 14.

3.8 The features "the second plurality of gear teeth" and "the first plurality of gear teeth" used in claim 14 have not been previously defined.

3.9 Claims 8 and 9 have been drafted as separate independent claims.

Under Article 84 in combination with Rule 43(2) EPC, an application may contain more than one independent claim in a particular category only if the subject-matter claimed falls within one or more of the exceptional situations set out in paragraph (a), (b) or (c) of Rule 43(2) EPC.

3.10 In the further prosecution of the application, failure to file an amended set of claims which complies with Rule 43(2) EPC, or to submit convincing arguments as to why the current set of claims does in fact comply with these provisions, may lead to refusal of the application under Article 97(2) EPC.

4 Novelty and Inventive Step (Art. 52(1), 54 and 56 EPC)

4.1 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claims 1 is not new within the meaning of Article 54(1) and (2) EPC.

Claim 1

D1 discloses (figures 2b, 6) a downhole rotational lock mechanism, comprising:

a tubular housing (23) having a longitudinal bore with an internal wall (see figures);

a driving gear (100) disposed in the longitudinal bore of the tubular housing, said driving gear having a peripheral edge secured to the internal wall of the longitudinal bore of the tubular housing (column 9 lines 26-33), said driving gear having an upper portion including a plurality of gear teeth (107) arranged around a central longitudinal bore through the driving gear (see figures);

a driven gear (101) movably disposed in the longitudinal bore of the tubular housing (to engage and disengage the clutch), said driven gear having a central longitudinal bore, said driven gear having a lower portion including a plurality of gear teeth (108); and

an output drive shaft (18) disposed longitudinally in the longitudinal bore of the tubular housing and in the longitudinal bore of the driven gear (see figures).

The subject-matter of claim 1 is therefore not new (Article 54(1) and (2) EPC).

4.2 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1, 8 and 9 does not involve an inventive step within the meaning of Article 56 EPC.

Claim 1

D2 is considered to be the prior art closest to the subject-matter of claim 1 and discloses (figures 1, 2, 3) a downhole rotational lock mechanism (25), comprising:

a tubular housing (11) having a longitudinal bore with an internal wall (see figures);

a driving gear (26) disposed in the longitudinal bore of the tubular housing, said driving gear having a peripheral edge secured to the internal wall of the longitudinal bore of the tubular housing (figure 2, page 3 lines 24-30), said driving gear having an upper portion including a ~~plurality of gear teeth~~ *lug* (29) arranged around a central longitudinal bore through the driving gear (see figures);

a driven gear (31) movably disposed in the longitudinal bore of the tubular housing (figures 2 and 3), said driven gear having a central longitudinal bore, said driven gear having a lower portion including a ~~plurality of gear teeth~~ *lug* (34); and

an output drive shaft (12) disposed longitudinally in the longitudinal bore of the tubular housing and in the longitudinal bore of the driven gear (see figures).

The subject-matter of claim 1 therefore differs from this known rotational lock mechanism in that it comprises a **plurality** of gear teeth.

The problem to be solved by the present invention may therefore be regarded as: how to increase the torque-transmitting capacity of rotational lock such as the one of D2.

The solution proposed in claim 1 of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC). Adding teeth on a torque-transmitting gear to increase its torque-transmitting capacity is a routine operation for the skilled person. Such designs are widely disclosed in the prior art (see for example document D3, figures 1 and 2).

Claims 8 and 9

D2 further discloses (page 3 lines 71-85):

rotating the tubular housing and the driving gear at a first rotational speed in a first rotational direction;

rotating the output shaft and the driven gear at a second rotational speed less than the first rotational speed and in the first rotational direction;

engaging the driven gear with the driving gear;

transferring rotational torque from the driving gear to the driven gear;

rotating the output shaft and the driven gear at a third rotational speed greater than the first rotational speed and in the first rotational direction;

disengaging the driven gear from the driving gear; and

discontinuing the transfer rotational torque from the driving gear to the driven gear.

The subject-matter of claims 8 and 9 is therefore not inventive (Articles 52(1) and 56 EPC).

4.3 Dependent claims 2-5, 7, 10-12 and 15-17 do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to novelty and/or inventive step.

Claims 2-5, 10-12 and 15-16

D2 discloses al the additional technical features of these claims (please refer to the relevant passages cited in the search report).

These claims are therefore considered not inventive.

Claims 7 and 17

D2 does not disclose a bias member provided to urge the driven gear toward the driving gear, as this is caused by gravity. However, bias members such as springs are widely-known in the industry to provide an additional force to urge an element toward another (see for example D3, figure 1). Such a feature is a basic constructional modification that would be included by the skilled person, as dictated by circumstances, without using inventive skills.

These claims are therefore not inventive.

5 Further Objections

5.1 The additional features of dependent claims 6, 13 and 14 are not disclosed in their present form in any of the documents cited in the search report. A new independent claim may be drafted to include these features, bearing in mind that the features known in combination in D1 and D2 should be placed in the preamble of such a claim in accordance with Rule 43(1) EPC.

5.2 The features of **all** the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion (see Guidelines F-IV, 4.19).

5.3 To meet the requirements of Rule 42(1)(b) EPC, D1 and D2 should be identified in the description and the relevant background art disclosed therein should be briefly discussed.

6 Procedure

6.1 The applicant is invited to file new claims which take account of the above comments.

6.2 In order to comply with the requirements of Rule 137(4) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see Guidelines H III, 2.1).

If the applicant considers it appropriate, these indications could be submitted in handwritten form on a copy of the relevant parts of the application as filed or in documents with tracked changes.

6.3 Amendments should be made by filing replacement pages. Unnecessary recasting of the description should be avoided. An amended abstract is not required. The applicant should also take account of the requirements of Rule

50(1) EPC. According to Rule 50(1) in conjunction with Rule 49(8) EPC, amendments shall be typed or printed. Handwritten amendments may only be made in documents other than those replacing application documents (Rule 50(2) EPC); they may, for example, be used to fulfil the requirements of Rule 137(4) EPC (identifying amendments and indicating basis for them).